INCH-POUND
MIL-DTL-32679
3 August 2021

DETAIL SPECIFICATION

CLOTH, TROPICAL, NYLON AND COTTON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers tropical, lightweight, nylon and cotton blend cloths.

1.2 Classification. The cloths are of the following types and classes (see 6.2).

1.2.1 Types.

Type I – Untreated

Type II – Treated

1.2.2 Classes.

Class 1 – Woodland Marine pattern (MARPAT)

Class 2 – Desert MARPAT

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-32659 - Textiles: Vector Protection Treatment of

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

16 CFR 303 - Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at www.ecfr.gov.)

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8305

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U.S. ARMY COMBAT CAPABILITIES DEVELOPMENT COMMAND SOLDIER CENTER DRAWINGS

2-1-2525 - Woodland MARPAT- 4 color (Coyote 476/498)

2-1-2526 - Woodland MARPAT- 4 color (Green 474 with EGA symbol)

2-1-2527 - Woodland MARPAT- 4 color (Black 477)

2-1-2528 - Woodland MARPAT- 4 color (Khaki 475)

2-1-2529 - Desert MARPAT- 4 color (Light Tan 479)

2-1-2530 - Desert MARPAT- 4 color (Urban Tan 478)

2-1-2531 - Desert MARPAT- 4 color (Light Coyote 481 with EGA symbol)

2-1-2532 - Desert MARPAT- 4 color (Highland 480)

(Copies of these drawings are available from the U.S. Army Combat Capabilities Development Command Soldier Center, ATTN: FCDD-SCP-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change

AATCC EP2 - Evaluation Procedure for Gray Scale for Staining

AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

AATCC TM8 - Test Method for Colorfastness to Crocking: AATCC Crockmeter Method

AATCC TM15 - Test Method for Colorfastness to Perspiration

AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon-Arc

AATCC TM20 - Test Method for Fiber Analysis: Qualitative

AATCC TM20A - Test Method for Fiber Analysis: Quantitative

AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated

AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles

AATCC TM135 - Test Method for Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at www.aatcc.org.)

ASTM INTERNATIONAL

ASTM D737 - Standard Test Method for Air Permeability of Textile Fabrics

ASTM D1424 - Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus

ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method

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ASTM D3775	-	Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	-	Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D3990	-	Standard Terminology Relating to Fabric Defects
ASTM D5034	-	Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at www.astm.org.)

CRC PRESS

Hayes, A. Wallace (editor) and Kruger, Claire L. (editor). *Hayes' Principles and Methods of Toxicology*. 2014.

(There are a number of ways through which this book may be obtained, such as online retailers, bookstores, and the public library, to name a few.)

OTHER PUBLICATIONS

Fabric Defect Replica Scales

(For information on obtaining copies of the Fabric Defect Replica Scales please contact U.S. Army Combat Capabilities Development Command (CCDC), Soldier Center, ATTN: FCDD-SCP-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

SAE INTERNATIONAL

AMS-STD-595 - Colors Used in Government Procurement

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.2).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Physical requirements. The cloth shall be a lightweight, tropical, nylon and cotton blend material and shall meet the requirements stated in table I when tested as specified in 4.6.

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TABLE I. Physical requirements.

Characteristic	Requirement
Nylon fiber	Continuous filament, round cross-section, bright luster, and high tenacity
Nylon yarn	140 denier
Cotton yarn	Ring spun, 36 single cotton count with a 3.75 “Z” twist factor
Yarn (warp and fill)	Nylon twisted with cotton singles yarn with a 3.75 “S” twist factor
Yarn blend	50±5 percent nylon with the remaining percentage cotton based on the dry weight of the desized cloth
Weave	Plain weave construction
Yarn count (minimum)	61 ends per inch in the warp and 51 picks per inch in the filling
Weight	4.8–5.8 ounces per square yard
Breaking strength (minimum)	
Warp	170 pounds
Filling	170 pounds
Tear strength (minimum)	
Warp	20 pounds
Filling	20 pounds
Seam strength (minimum)	110 pounds
Air permeability (minimum)	50 cubic feet per minute
Dimensional change	
Shrinkage (warp and filling)	≤3.5 (individual unit), ≤3.0 (lot average)
Growth (warp and filling)	0 (individual unit), 0 (lot average)
Visual shade matching	
Daylight D75 (7500±200K) (with illumination of 100±20 foot candles)	Good match to standard sample
Incandescent (2856±200K)	Good match to standard sample
Camouflage pattern execution	Good match to standard sample (see 3.4.2)
Colorfastness (minimum)	
Light, after 40 AATCC Fading Units (AFU) or 170 kJ/(m^2nm) 1/ at 420 nm	4
Perspiration (acid and alkaline)	4
Laundering (after 4 cycles)	
Color change	4
Staining	
All colors (except black 477)	3–4
Black 477	2–3

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TABLE I. Physical requirements – Continued.

Characteristic	Requirement
Crocking (wet and dry)	
All colors (except black 477)	3.5
Black 477	1.5
Spectral reflectance	See 3.4.3
pH (water extract)	5.0–8.5
Labile sulfur content	See 3.4.4
FOOTNOTE: 1/ Kilojoules (kJ) per square meter per nanometer (m^2nm).	

3.4.1 Camouflage print. The camouflage print shall be as specified in the contract (see 6.2). The camouflage pattern shall be roller or screen printed. Resin-bonded pigments are not permitted except for a small amount of carbon black pigment, which may be used to meet the black shade providing all other requirements are met. The cloth shall conform to the requirements of table II or table III below when tested as specified in 4.4, 4.5, and 4.6.

TABLE II. Class 1 Woodland MARPAT camouflage pattern requirements.

Characteristic	Requirement
Ground shade (dyed)	Khaki 475 or color chip 23420 of AMS-STD-595
Camouflage pattern (printed)	Drawing 2-1-2525
	Drawing 2-1-2526
	Drawing 2-1-2527
	Drawing 2-1-2528
Color	Green 474 or color chip 24112 of AMS-STD-595
	Khaki 475 or color chip 23420 of AMS-STD-595
	Coyote 476/498 or color chip 20150 of AMS-STD-595
	Black 477 or color chip 37029 of AMS-STD-595

TABLE III. Class 2 Desert MARPAT camouflage pattern requirements.

Characteristic	Requirement
Ground shade (dyed)	Light Tan 479 or color chip 23530 of AMS-STD-595
Camouflage pattern (printed)	Drawing 2-1-2529
	Drawing 2-1-2530
	Drawing 2-1-2531
	Drawing 2-1-2532
Color	Light Tan 479 or color chip 23530 of AMS-STD-595
	Urban Tan 478 or color chip 20270 of AMS-STD-595
	Light Coyote 481 or color chip 20220 of AMS-STD-595
	Highland 480 or color chip 20090 of AMS-STD-595

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3.4.2 Camouflage pattern execution. The camouflage pattern shall reproduce the standard sample with respect to design, colors, and registration of the respective areas when tested as specified in 4.4, 4.5, and 4.6. The pattern repeat of the camouflage-printed finished cloth shall be 35±1½ inches. Each pattern area shall show solid coverage. Skitteriness, feathering, grinning, haloing, or trapping shall not exceed that shown on the standard sample. When the standard sample is not referenced for pattern execution, pattern drawings will be provided, and the pattern of the finished cloth shall match that of the drawings.

3.4.3 Spectral reflectance.

3.4.3.1 Class 1. The cloth shall meet the spectral reflectance values (in percent) for the visible/near infrared wavelength range, 600 to 860 nanometers, for the colors specified in table IV, as applicable, when tested as specified in 4.6.2.1.

TABLE IV. Class 1 spectral reflectance values (percent).

Wavelengths	Black 477		Coyote 476/498 & Khaki 475		Green 474	
Nanometers	**Min.**	**Max.**	**Min.**	**Max.**	**Min.**	**Max.**
600	--	10	8	18	3	10
620	--	10	8	18	3	10
640	--	10	8	18	3	9
660	--	10	8	18	3	12
680	--	10	10	22	3	14
700	--	10	18	33	5	18
720	--	10	22	45	7	20
740	--	10	30	55	12	28
760	--	10	35	65	18	36
780	--	10	40	75	26	44
800	--	10	45	80	34	52
820	--	10	50	86	42	60
840	--	10	55	88	53	68
860	--	10	60	90	56	74

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3.4.3.2 Class 2. The cloth shall meet the spectral reflectance values (in percent) for the visible/near infrared wavelength range, 700 to 860 nanometers, for the colors specified in table V when tested as specified in 4.6.2.1.

TABLE V. Class 2 spectral reflectance values (percent).

Wavelengths	Light Tan 479		Light Coyote 481 & Highland 480		Urban Tan 478	
Nanometers	Min.	Max.	Min.	Max.	Min.	Max.
700	38	53	19	41	25	44
720	38	54	20	41	25	45
740	39	55	20	42	25	46
760	40	56	21	42	26	47
780	41	57	21	42	27	48
800	43	58	22	43	28	50
820	45	59	23	45	30	52
840	48	62	24	46	33	55
860	50	65	25	48	36	58

3.4.4 Labile sulfur. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid shall be chosen and applied such that the dyed and finished cloth shall show no more than a rating of "slight" labile sulfur when tested as specified in 4.6.2.2.

3.5 Insect repellent treatment (type II only). The type II cloth shall be treated with an insect repellent treatment conforming to the requirements of MIL-PRF-32659, except that the active ingredient concentration and mosquito percent bite protection shall conform to tables VI and VII when tested as specified in 4.7.

TABLE VI. Insect repellent treatment level requirements.

Repellent active ingredient (AI)	Condition	AI concentration (mg/cm^2)	
		Minimum	Maximum
Etofenprox	Initial	0.132	0.195
Permethrin	Initial	0.076	0.112

TABLE VII. Mosquito percent (%) bite protection.

	Percent bite protection
Initial	80% minimum, 96% objective
After 20 launderings	65% minimum, 96% objective
After 50 launderings	50% minimum, 90% objective

3.6 Toxicity. The finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.8. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

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3.7 Length, width, and put-up.

3.7.1 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards, and each length shall be put-up on a roll when examined in accordance with 4.4.3.

3.7.2 Width. For Government procurements only, unless otherwise specified (see 6.2), the width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.8 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with 16 CFR 303 when examined as specified in 4.4.4.

3.9 Insect repellent treatment roll identification (type II only). Each roll of type II cloth shall be labeled in accordance with MIL-PRF-32659.

3.10 Workmanship. The cloth shall conform to the quality of product established by this specification, be uniform, and be free from defects specified in 4.4 and those defects that adversely affect form, fit, or function. The demerit points per 100 square yards, when calculated as specified in section 4, shall not exceed the applicable established maximum point values (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

3.11 Disposal of rejected items. Rejected printed seconds and mill seconds shall be disposed of as specified in the contract (see 6.2 and 6.6.1).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1.1, shall be examined and tested in accordance with 4.4 through 4.7 for compliance with the appearance, color, finish, design, configuration, workmanship, and testing requirements specified in this document.

4.2.1 First article samples and acceptance criteria. Unless otherwise specified in the contract (see 6.2), a first article sample shall be provided; the sample unit shall be one continuous 5-yard length of full width cloth; and the presence of excessive defects or failure of any testing shall be cause for rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified in the contract (see 6.2), conformance inspection, in accordance with 3.1.2, shall consist of the examinations and tests as specified in 4.4 through 4.7. The government reserves the right to inspect and test all components, materials, and end items to determine conformance to stated requirements.

4.4 Yard-by-yard examination. Unless otherwise specified in the contract (see 6.2), the sample size shall be 20 rolls selected from 20 containers. Each roll in the sample shall be examined yard-by-yard on the printed side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. The lot shall be unacceptable if the points per 100 square yards examined exceeds 20.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 30.0 points. If one roll in the sample exceeds 30.0 points per 100 square yards, a second sample of 20 rolls shall be examined for individual roll quality only. The lot shall be unacceptable if one or more rolls in the second sample exceeds 30.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{Total\ points\ scored\ in\ sample\ \times\ 3600}{Contracted\ width\ of\ cloth\ (inches)\ \times\ Total\ \ yards\ inspected} = Points\ per\ 100\ square\ yards$$

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4.4.1 Defects. Defects, as defined in ASTM D3990, include, but are not limited to, abrasion marks, smash, missing or broken yarns, visible mends, knots, thin place, dye streaks, or misweaves affecting appearance or serviceability. All defects that are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.4.2, except as follows:

a. Only coarse yarns that exceed twice the normal diameter shall be scored.

b. Mixed filling shade bar shall be scored only when resulting from wrong ply, wrong twist in the yarn, or off-shade yarn.

c. Only knots and slubs that exceed limits shown on Fabric Defect Scale (see 2.3) “F” for slubs and “D” for knots shall be scored.

The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points.

4.4.2 Demerit points. Demerit points shall be assigned as follows in table VIII. All defects as specified in table IX shall be scored four points for each yard in which they occur.

TABLE VIII. Demerit points by defect size.

Defect size	**Point value**
Up to 3 inches in any dimension	One point
Exceeding 3 inches, but not exceeding 6 inches in any dimension	Two points
Exceeding 6 inches, but not exceeding 9 inches in any dimension	Three points
Exceeding 9 inches in any dimension	Four points

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TABLE IX. Defects.

#	Defect description
1	Any hole, cut, pinhole, tear, scratch, or abrasion mark 1/.
2	Edge ravels when pulled outward.
3	Uneven weaving throughout.
4	Camouflage pattern execution (design, colors, and registration) not equal to the standard sample.
5	Skitteriness (mottled, uneven color) of pattern exceeds that shown by the standard sample.
6	Feathering or spew (fuzziness at color boundaries) of pattern exceeds that shown by the standard sample.
7	Grinning (off register, gap where ground shade shows through) of pattern exceeds that shown by the standard sample.
8	Haloing or trapping (overlapping of colors) of pattern exceeds that shown by the standard sample.
9	Tackiness (sticky to touch).
10	Any spot, stain, or streak more than 1 inch in combined directions. 2/
11	Any objectionable odor. 3/
12	Width not as specified within a tolerance of ±½ inch.
13	Broken or missing yarn. 2/
14	Any embedded foreign matter or uncleanness. 2/
15	Slack or tight selvages. 4/
16	Edges rolled, folded, scalloped, or corded, or baggy, ridgy, or wavy cloth.

FOOTNOTES:

1/ The same yardage shall be given a through-lighting inspection for pinholes and thinly coated areas.

2/ Clearly visible at normal inspection distance (3 feet) by either direct viewing or through-lighting.

3/ Odors of chemicals commonly used in coating compounds shall not be considered objectionable.

4/ To determine the presence of unacceptable selvage conditions, the following procedure shall be observed: During the visual examination, the perch shall be stopped a minimum of three times for each roll in the sample, the tension removed, and the finished cloth examined for the selvage conditions. A waviness in the selvage or significant ripples diagonally across the width of the fabric is an indication of slack or tight selvages.

4.4.3 Length and put-up examination. Unless otherwise specified in the contract (see 6.2), each roll in the sample shall be examined for length during the yard-by-yard examination. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined to ensure ticketing or labeling compliance with 16 CFR 303. The lot shall be unacceptable if two or more rolls are not in compliance.

4.5 Verification of shade. Unless otherwise specified in the contract (see 6.2), the cloth shall be examined for color, pattern execution, and uniformity of shade.

4.5.1 Sampling for verification of shade. The finished cloth shall be 100 percent sampled for shade. A 16-inch by full width swatch (header) will be cut from each roll in the lot.

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4.5.2 Swatch identification. Each swatch shall be identified with the following information:

a. "FOR GOVERNMENT SHADE APPROVAL"
b. Name of prime contractor
c. Contract number
d. Nomenclature
e. Government inspection lot number
f. Number of swatches being submitted
g. Quality Assurance Representative (QAR)'s name
h. Name and address of the QAR's base plant
i. Piece number of standard sample cited
j. Tally list with piece numbers

4.5.3 Shade evaluation. Each submitted swatch shall be evaluated for shade against the standard sample. If a swatch is rejected for shade, the roll from which the rejected swatch was cut shall be removed from the lot. If a swatch is accepted for shade, the yardage on the roll shall be adjusted to remove the sample length.

4.5.4 Reworks and resubmitted pieces. If the contractor reworks and resubmits yardage originally rejected for shade or finish, the contractor shall not combine the reworked yardage with normal production or with lots rejected for other causes. Such yardage shall be combined to form one resubmitted lot. Each roll shall retain its original piece number, suffixed with an "x".

4.6 Physical requirements verification. Unless otherwise specified in the contract (see 6.2), the cloth shall be examined for physical requirements.

4.6.1 Sampling and acceptance criteria for verification of physical requirements. Sampling shall be in accordance with table X. The sample unit shall be five continuous yards full width of the finished cloth per roll of cloth in the sample. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified in this document.

TABLE X. Physical requirements verification sampling.

Lot size	Sample size (# of sample units)
800 or less	2
801 through 22,000	3
22,001 and over	5

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4.6.2 Physical requirements testing. The cloth shall be examined and tested in accordance with the specified examinations and tests referenced in table XI.

TABLE XI. Physical requirements testing.

Characteristic	Requirement	Test Method
Fiber identification	3.4	AATCC TM20 1/
Nylon fiber	3.4	AATCC TM20 and visual 1/
Nylon yarn	3.4	ASTM D1907/1907M 1/
Cotton yarn	3.4	1/
Yarn twist	3.4	Visual 1/
Yarn blend	3.4	AATCC TM20A 1/ 2/
Weave	3.4	Visual 1/
Yarn count	3.4	ASTM D3775 1/
Weight	3.4	ASTM D3776/D3776M 1/
Breaking strength		
Warp	3.4	ASTM D5034 3/
Filling	3.4	ASTM D5034 3/
Tear strength	3.4	
Warp	3.4	ASTM D1424 3/
Filling	3.4	ASTM D1424 3/
Seam strength	3.4	ASTM D5034 3/
Air permeability	3.4	ASTM D737 3/
Dimensional change	3.4	
Shrinkage (warp and filling)	3.4	AATCC TM135, 3, V, Aiii
Growth (warp and filling)	3.4	AATCC TM135, 3, V, Aiii
Visual shade matching		
Daylight D75 (7500±200K) (with illumination of 100±20 foot candles)	3.4	AATCC EP9, option C
Incandescent (2856±200K)	3.4	AATCC EP9, option C
Camouflage pattern execution	3.4	AATCC EP9, option C
Colorfastness (minimum)		
Light (after 40 hours or 170 kilojoules)	3.4	AATCC TM16.3, option 3 and AATCC EP1
Perspiration (acid and alkaline)		
Color change	3.4	AATCC TM15 and AATCC EP1
Staining		AATCC TM15 and AATCC EP2
Laundering (after 4 cycles)		
Color Change	3.4	AATCC TM61 test 1A 4/ and AATCC EP1
Staining	3.4	AATCC TM61 test 1A 4/ and AATCC EP2
Crocking	3.4	AATCC TM8 and AATCC EP8

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TABLE XI. Physical requirements testing – Continued.

Characteristic	Requirement	Test Method
Spectral reflectance	3.4	See 4.6.2.1
pH (water extract)	3.4	AATCC TM81
Labile sulfur content	3.4	See 4.6.2.2

FOOTNOTES:

1/ See 6.8.

2/ Cotton content shall be calculated as follows: Cotton content, percent = R/S x 100 where R= Weight of residual fibers and S= Weight of dry desized specimen.

3/ Five determinations required.

4/ Except 1993 AATCC Standard Reference Detergent (non-phosphate) without optical brightener shall be used.

4.6.2.1 Spectral reflectance. The cloth shall meet the spectral reflectance requirements specified in 3.4.3.1 and 3.4.3.2. Spectral reflectance data shall be determined on the face side and shall be obtained from 600 to 860 nanometers for class 1 and from 700 to 860 nanometers for class 2 at 20-nanometer intervals on a spectrophotometer relative to a polytetrafluoroethylene (PTFE) family of compounds, the preferred white reference standard. Other white reference materials may be used, provided they are calibrated to absolute white or vitrolite tiles. The spectral bandwidth shall be less than 26 nanometers at 860 nanometers. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a source that simulates either International Commission on Illumination (CIE) source A or CIE source D65. The specimen shall be measured as a single layer, backed with six layers of the same fabric and shade. Measurements shall be taken on a minimum of two different areas and the data averaged. The measured areas should be taken at least 6 inches away from the selvage. The specimen shall be viewed at an angle no greater than 10 degrees from the normal, with the specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy within 2 nanometers. The standard aperture size used in the color measurement device shall be 0.3725 inch in diameter. Any color having spectral reflectance values outside the limits at four or more of the wavelengths specified shall be considered a test failure.

4.6.2.2 Presence of labile sulfur test. The cloth shall meet the requirements specified in 3.4.4. In the determination of the presence of labile sulfur in textile materials with lead acetate, two 1.50±0.01-gram samples from each material submitted for evaluation shall be tested. Each of the two samples shall be cut into very small pieces and placed into separate test tubes. The samples shall be submersed into a stannous chloride solution that contains 100 grams of American Chemical Society (ACS) grade stannous chloride crystals in 100 milliliters of ACS grade hydrochloric acid (35 percent concentration) and 50 milliliters of distilled water. A filter paper wet out with a 5.0 percent lead acetate solution shall be placed over the top of the test tube. The lead acetate solution contains 5.0 grams of chemically pure (CP) reagent grade lead acetate and enough distilled water to make up a 100-milliliter solution; if the solution is not clear, a few drops (one at a time) of glacial acetic acid shall be added until the solution is clear. The test tube containing the textile sample, stannous chloride, and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown or black stain on the filter paper shall be evaluated as follows:

a. Free – The filter paper shows no discoloration or staining of any kind.
b. Slight – The filter paper shows a light tan to light brown discoloration stain.
c. Moderate – The filter paper shows a dark brown discoloration stain.
d. Severe – The filter paper shows a black color stain.

Any rating other than “Free” or “Slight” shall be considered a test failure.

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4.7 Insect repellent treatment verification (type II only). Unless otherwise specified (see 6.2), verification of the insect repellent treatment shall be in accordance with MIL-PRF-32659. The contractor shall notify and request Government approval any time there is a change in the treatment formulation or processing conditions. Changes shall include manufacturing supply chain, active ingredient, and application method. Insect repellent active ingredient treatment level and percent mosquito bite protection shall be verified before the Government approves the change (see 6.2).

4.8 Toxicity test. When specified (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see Hayes' Principles and Methods of Toxicology). If toxicity requirement (see 3.6) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the Marine Corps Tropical Combat Utility Uniform (MCTCU) blouse and trousers for wear by military personnel of the United States Marine Corps in hot weather combat environments. Type I cloth may be utilized in blouses and trousers that are treated with insect repellent after the item has been cut and sewn. Class 1 is for wear in woodland environments, and class 2 is for wear in desert environments.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Types and classes required (see 1.2).
c. National stock numbers (NSNs).
d. When first article is required (see 3.1.1 and 6.4).
e. When conformance inspection is required (see 3.1.2 and 4.3).
f. Applicable Government standard samples and drawings (see 3.2 and 3.4.1).
g. Length, width, and put-up required (3.7.1 and 3.7.2).
h. Maximum acceptable point values (see 3.10).
i. Requirement for the disposal of rejected printed seconds and mill seconds (see 3.11 and 6.6.1).
j. First article sampling and acceptance criteria (see 4.2.1).
k. Yard-by-yard examination (see 4.4).
l. Length and put up (see 4.4.3).
m. Shade examination (see 4.5).
n. Physical requirements verification (see 4.6).
o. Insect repellent treatment verification (see 4.7).

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p. When toxicity testing is required (see 4.8).

q. Packaging requirements, including insect protection labeling for type II cloth (see 5.1 and 6.7).

r. A certificate of compliance, if acceptable for physical requirements (see 6.8).

6.3 Information requests. For access to information such as specifications, drawings, and standard shade samples of cloth, go to https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx for the Defense Logistics Agency Troop Support's Clothing and Textile Specification/Pattern/Drawing request form. Complete the request form and then submit the form. Request to use equivalent materials or components should be sent to the contracting officer for approval by the military service.

6.4 First article. When first article inspection (see 3.1.1) is specified (see 6.2), the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.2.1. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government and that bidders offering such products who wish to rely on such a production or test must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not alternate submit bids unless specifically requested to do so in the solicitation.

6.5 Replacement document (in part). This document replaces the lightweight nylon and cotton basic material requirements previously stated in MIL-PRF-MCTCU dated APR 2020.

6.6 Government intellectual property and trademark rights notification. The Government possesses intellectual property/trademark rights in the following Marine Corps patterns and logos (collectively referred to as "intellectual property"): the Eagle, Globe, and Anchor (EGA) logo, including the EGA logo as it appears embedded in the fabric pattern. The Government further has title to the invention disclosed and claimed in United States Design Patent No. D491,372 issued on 15 June 2004 for "Camouflage Pattern for Sheet Material and Uniforms" and the invention disclosed and claimed in United States Patent No. 6,805,957 issued on 19 October 2004 for "Camouflage U.S. Marine Corps Utility Uniform: Pattern, Fabric, and Design." The Government claims exclusive ownership of the above mentioned intellectual property; therefore, no entity other than the Government, or those contracted by or having obtained proper permission or licenses from the Government to do so, are permitted to produce, sell, or transfer in any manner any items (clothing or non-clothing) containing or copying, in whole or in part, the intellectual property. Doing so will be considered an infringement on the Government's intellectual property rights and will be subject to legal action.

6.6.1 Disposal of types I and II cloth. For types I and II, all scraps, seconds, irregulars, extra material, and garments containing the aforementioned intellectual property/trademarks that are not utilized for Government contracts or a purpose authorized in writing by the Government are to be destroyed and not sold or transferred in any manner. This restriction applies to the prime contractor, as well as all subcontractors, and is to be incorporated into all agreements with subcontractors (see 6.2).

6.7 Type II (insect-repellant-treated) packaging. Unless otherwise specified (see 6.2), packaging of the type II cloth should be in accordance with MIL-PRF-32659.

6.8 Certificates of compliance. Certificates of compliance may be accepted for selected requirements (see 6.2). It is recommended that vendors verify acceptability prior to using certificates of compliance to address requirements.

6.9 Subject term (key word) listing.

Blouse

Camouflage

Combat

Desert

Fabric

Insect repellent

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Jungle
Marine Corps
MCTCU
Trouser
Uniform
Woodland

CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC
Air Force – 11

Review activities:
Army – AV, CR, IH, MI
Navy – AS, CG1, NU
Air Force – 03
DLA – CT

Civilian agency:
GSA – FAS

Preparing activity:
Navy – MC
(Project 8305-2020-028)

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